

FIELD THEORY PART-II

Phase 4: Stokes' Theorem and Ampere's Law

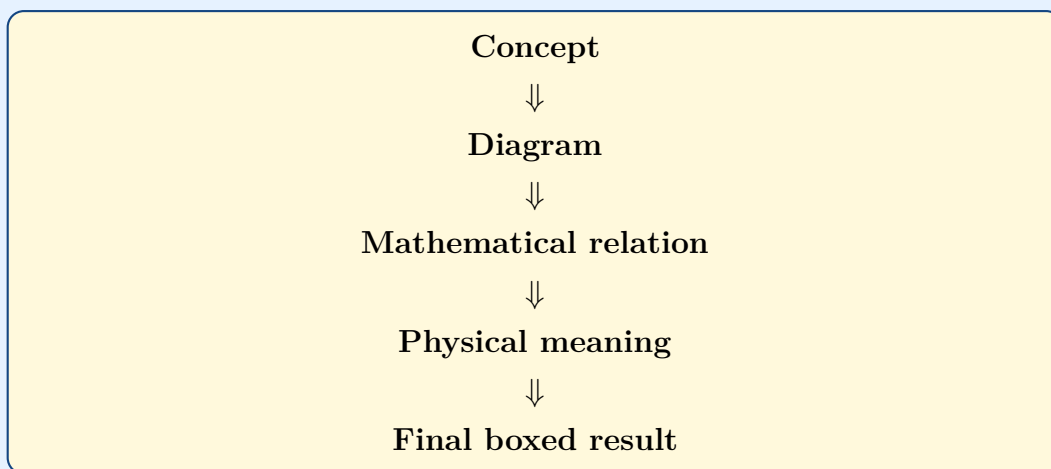
From circulation theorem to magnetostatic differential law

Academic Article Style Notes for Semester Preparation

How to Use This Phase

This phase is made for semester preparation from a very basic level. Read the concept first, then understand the diagram, then learn the derivation pattern, and finally practise the exam-style writing block.

The recommended writing order is:



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1 Aim of Phase 4

Main Goal

This phase connects mathematics with magnetostatics. The main targets are Stokes' theorem and Ampere's law:

$$\oint_C \mathbf{A} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

and

$$\nabla \times \mathbf{H} = \mathbf{J}$$

2 Stokes' Theorem

Statement

Stokes' theorem states that line integral of a vector field around a closed boundary is equal to surface integral of curl over the surface bounded by that boundary.

$$\oint_C \mathbf{A} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

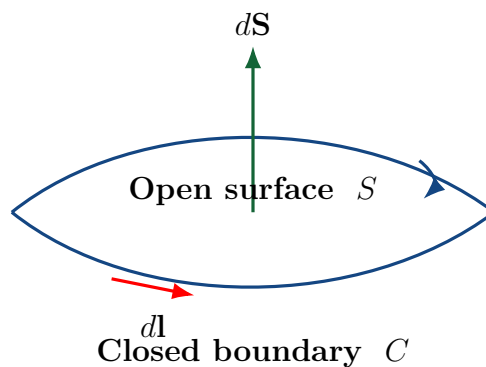


Figure 1: Stokes' theorem connects boundary circulation with surface curl.

3 Idea Behind Stokes' Theorem

Internal Boundary Cancellation

Divide the surface into many small surfaces. Each small surface has its own circulation. When all small circulations are added, internal paths cancel in opposite pairs. Only the outer boundary remains. Therefore, total surface curl equals boundary circulation.

4 Ampere's Circuital Law

Integral Form

Ampere's circuital law is:

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}$$

The enclosed current can be written as:

$$I_{\text{enc}} = \iint_S \mathbf{J} \cdot d\mathbf{S}$$

5 Derivation of Differential Form

Using Stokes' Theorem

From Ampere's law:

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \iint_S \mathbf{J} \cdot d\mathbf{S}.$$

By Stokes' theorem:

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}.$$

Therefore:

$$\iint_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} = \iint_S \mathbf{J} \cdot d\mathbf{S}.$$

Since the surface is arbitrary:

$$\nabla \times \mathbf{H} = \mathbf{J}$$

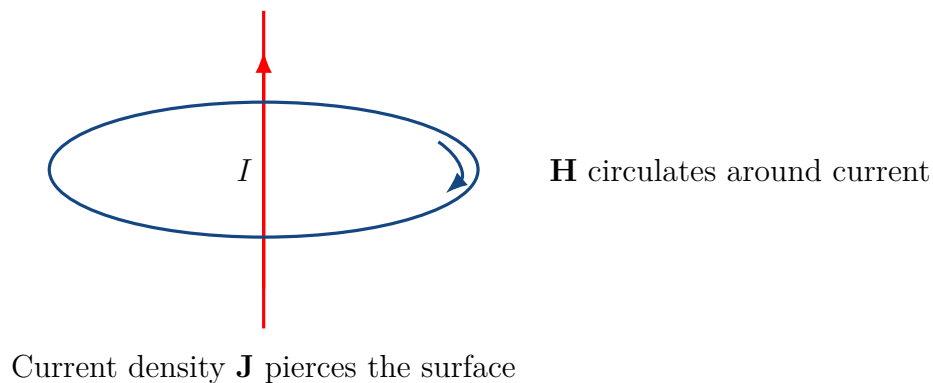


Figure 2: Physical meaning of $\nabla \times \mathbf{H} = \mathbf{J}$.

6 Physical Significance

Meaning

$$\nabla \times \mathbf{H} = \mathbf{J}$$

means current density is the cause of circulation of magnetic field intensity.

7 Exam Writing Template

For Stokes' Theorem

Write:

1. Statement.
2. Diagram of open surface and closed boundary.
3. Explain small surface division.
4. Internal boundaries cancel.
5. Final boxed theorem.

For Ampere Law

Write:

1. Ampere's circuital law.
2. Write current as surface integral of $\mathbf{J}$.
3. Apply Stokes' theorem.
4. Compare integrands.
5. Final result and physical meaning.

8 Final Self-Test for This Phase

Instructions

Answer these without looking at the notes. If you cannot answer a point, revise that section again before moving to the next phase.

1. Can I state the main definition of this phase?
2. Can I draw the main diagram neatly without overlap?
3. Can I explain the physical meaning in two or three lines?
4. Can I write the final formula with correct symbols?
5. Can I connect this topic with at least one past-year question pattern?
6. Can I write a semester-style answer with definition, diagram, derivation and final result?

Completion Rule

This phase is complete only when the concept, diagram, formula and answer-writing pattern are all ready.